

Assignment – 2

Solve the problems

Lecture – 4:

1. State and explain the Ampere's law. What is the critical form of it?
2. Drive the equation of the magnetic field for a long straight current carrying wire.
3. Formulate the magnitude of the magnetic field of a long solenoid.
4. Draw a cross-sectional view of a toroid and write down the equation of the magnetic field of it.
5. Define the hall effect and hall voltage. What are the applications of it?
6. A solenoid is $1.0m$ long and $3.0cm$ in mean diameter. It has five layers of windings of 850 turns each and carries current of $5.0amp$. (a) What is B at its centre? (b) What is the magnetic flux ϕ_B for a cross-section of the solenoid at this centre? Ans: a) $2.67 \times 10^{-2}T$ b) $1.88 \times 10^{-5}wb$
7. An air-core solenoid with 2000 loops on it is $600mm$ long and has diameter of $20mm$. If a current of $5A$ is sent through it, what will be the flux density within? Ans: $2.09 \times 10^{-2}T$
8. A solenoid $0.5m$ long has 2000 turns. The magnetic induction near the centre of the solenoid is $0.08T$. What is the current in the solenoid? Ans: $15.92A$
9. A diameter of a circular coil is $31.4 \times 10^{-2}m$ and its number of turns is 400. For what amount of current flow in the coil, the magnetic field at the center of the coil will be $8 \times 10^{-10}wbm^{-2}$? Ans: $5 \times 10^{-7}A$
10. $5A$ current is flowing through a conducting wire of length of $1m$. Calculate the magnetic field at a distance $5cm$ from the wire. Ans: $2 \times 10^{-5}T$
11. The length and diameter of a solenoid are respectively $2m$ and $2cm$. It has 10 layers. Each layer consists of 1000 turns. Determine the field intensity and flux at the center of the solenoid of $5A$ current flows in it. Ans: a) $3.14 \times 10^{-2}T$ b) $9.85 \times 10^{-6}wb$
12. The radius of the circular coil is $0.02m$ and its number of turns is 100. Calculate the magnetic field at the center of the coil if $5A$ current flows in the coil. Ans: $0.0157T$

Lecture – 5:

1. What is the electromagnetic induction?
2. State and explain Faradays law of inductions. Write down the mathematical form of Newman's law.
3. State the Lenz's law.
4. Explain the energy conservation in electromagnetic induction.
5. Write a short note on: i) Self-induction ii) Mutual induction
6. Define 1 henry.
7. Explain the basic construction transformer.
8. Magnetic flux of $3 \times 10^{-8}wb$ crosses through a coil having 500 turns. The flux decreases to $3 \times 10^{-3}wb$ in 0.015 . What is the induced $e.m.f.$?
9. When the current of $2A$ flows through a coil of 400 turns creates magnetic flux of $4 \times 10^{-4}wb$. Calculate the self – inductance of the coil.
10. Magnetic flux in a coil of 400 turns changes from $8 \times 10^{-5}wb$ to $50 \times 10^{-5}wb$ in $3 \times 10^{-3}sec$. Calculate the induced emf in the coil.
11. Average emf induced in a coil is $32V$ when current decreases from $10A$ to $2A$ in $0.1sec$ in the coil. Calculate the self – inductance in the coil.
12. If the current in the primary coil is changed from $6A$ to $1A$ in $0.05sec$, then emf $5V$ is induced in secondary coil. What is the mutual inductance of the coil?
13. Two nearby coil A and B have number of turns respectively of 200 and 1000. When $2A$ current flows through the coil, then magnetic flux $2.4 \times 10^{-4}wb$ is produced in the coil A and $1.6 \times 10^{-4}wb$ in the coil B. Calculate (i) the self – inductance of the coil A (ii) the mutual inductance of the coil B. (iii) If current stops A in $0.4 sec$, find the induced emf in coil B.

Lecture – 6:

1. Draw the graphical representation of AC and DC.
2. Define form factor of AC.
3. The peak value of an alternating current is $20A$ and its frequency is $50hz$. Find the root mean square value of the current. How long will take to reach the peak from the zero?
4. The amplitude of an AC source is $160V$ and its frequency is $60Hz$. If a resistance of 20Ω is connected with the source, find the effective voltage, effective current and loss of energy due to heat per time.
5. If the equation of the alternating current is $i = 50 \sin 628t$, then find (a) peak value (b) frequency and (c) rms value (d) average value of the current.
6. The alternating emf is expressed as $E = 200 \sin 100\pi t$. What is its amplitude, average value, rms value and frequency?

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